COM5231 · Introduction to Wireless and Ad Hoc Networking

Part 6 — Model Answers

Test 1B · Test 2 2025 · Test 2 2026 · Bluetooth Exercises · Error-Detection Worksheet

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

Part 6 — Model Answers: Test 1B, Test 2 2025, Test 2 2026, Bluetooth Exercises and Error-Detection Worksheet

Conventions as in Part 5. Where a question is identical to one already answered in Part 5, the answer says so and gives any extra depth the higher mark allocation needs.

COM5231 TEST 1, VERSION B (T1B; 100 marks)

Q1. Explain the difference between single-bit and burst errors and the extent to which parity checking is appropriate as an error-detecting technique in both cases. (10)

Definitions SUPPLEMENTARY (standard; the worksheet [ED] lists this question but gives no theory). Causes from [WC-B].

Single-bit error (2 marks): **only one bit** of a data unit (byte, frame) is changed, from 0 to 1 or 1 to 0. E.g. 0000 0010 sent, 0000 1010 received. These are relatively **rare** in practice, because noise usually lasts longer than one bit time.

Burst error (3 marks): **two or more bits** in the data unit are changed. They need not be consecutive. The **burst length** runs from the **first** corrupted bit to the **last**, inclusive. Bursts are the **common case** in wireless: **impulse noise** (*"the primary source of error in digital data... a 0.01 s spike... would wash out about 10 000 bits at 1 Mbps"* [WC-B 5.2]) and **fading** dips of 20–30 dB [WC-B 5.3] corrupt many consecutive bits.

Parity checking (even parity: one bit appended so the number of 1s is even; the receiver recounts):

- **Single-bit errors (2 marks)**: parity **always detects** them. One flipped bit changes the count of 1s from even to odd. It is **cheap** (one extra bit) and simple, so it is **appropriate** where errors are isolated. It **cannot correct** them, though: it can't tell **which** bit flipped, so retransmission is needed.
- **Burst errors (3 marks)**: parity detects a burst **only if an odd number of bits** is flipped. Any **even** number of errors (2, 4, ...) leaves the parity unchanged and goes **undetected**. E.g. 10110010 → 1**10**10010: two errors, still four 1s, so it passes. Roughly **half of all bursts escape**, so single parity is **not appropriate** for bursty wireless channels. **Two-dimensional parity** improves this (it detects many bursts and corrects single errors), but **CRC** is preferred, since it detects **all bursts of length ≤ r** (the number of check bits).

Q2. Transmit the message 10011010 protected by the CRC polynomial $x^3 + x^2 + 1$.

(a) Use polynomial long division to determine the message that should be transmitted. (10)

1. **Generator**: $x^3 + x^2 + 0{\cdot}x + 1$ → **1101**; degree r = **3**.
2. **Append 3 zeros**: $M(x){\cdot}x^3$ → **10011010 000**.
3. **Divide by 1101 (modulo-2)**:

Step	Leading window	XOR 1101	Result	Remaining dividend
1	1001	1101	0100	0100 1010 000
2	1001	1101	0100	0010 0010 000
3	1000	1101	0101	0001 0110 000
4	1011	1101	0110	0000 1100 000
5	1100	1101	0001	0000 0001 000
6	1000	1101	0101	0000 0000 101

```
1 0 0 1 1 0 1 0 0 0 0
1 1 0 1
-------
  1 0 0 1
  1 1 0 1
  -------
    1 0 0 0
    1 1 0 1
    -------
      1 0 1 1
      1 1 0 1
      -------
        1 1 0 0
        1 1 0 1
        -------
              1 0 0 0
              1 1 0 1
              -------
                1 0 1   = remainder
```

4. **Remainder (CRC) = 101.** Polynomial check: $x^{10} + x^7 + x^6 + x^4 \bmod (x^3 + x^2 + 1) = \mathbf{x^2 + 1} \leftrightarrow 101$ ✔.
5. **Transmitted message = 10011010 101**. Check: 10011010101 ÷ 1101 → remainder **000** ✔.

(b) The leftmost bit is inverted by noise. What is the result of the receiver's CRC calculation? How does the receiver know an error occurred? (6)

- Received: **0** 0011010101 = **00011010101**.
- The receiver divides by 1101. The leading zeros need no XOR; the first 1 is at position 4: 1101 ⊕ 1101 = 0000 → **00000000101**.
- **Remainder = 101 (≠ 000).**
- A correctly received frame is **exactly divisible by G** (remainder 0), because the sender made it so. A **non-zero remainder** therefore proves the frame was corrupted. The receiver **detects the error**, discards the frame and requests retransmission (ARQ). (This matches the worksheet's own example: *"Since the remainder is non-zero, we know that something went wrong!"* [ED].)

Q3. Explain the three basic propagation mechanisms in a mobile communication system. (12)

Identical to T1A Q1 (Part 5).

Q4. Differences between path loss, multipath propagation and interference. (12)

Use T1A Q3 (Part 5): the Exam 1(a) paragraphs + table + the extra points. That gives 4 marks per impairment.

Q5. Discuss three methods to increase the capacity of a wireless communication system. (12)

Capacity = how many users or bits/s a system can carry in its spectrum. Three methods (4 marks each):

1. Frequency reuse with smaller cells (network densification / SDMA)

- Limit each transmitter's **range** and **reuse the same frequencies** in cells far enough apart that interference is acceptable. Traditional **macro cells** (km wide) are complemented by **small cells**: indoor **femtocells**, outdoor **picocells/relays**. Every reuse multiplies capacity, hence "network densification" [WC-B 5.4; WC-S 52].
- **SDMA/sectorization** strengthens this: directional antennas reduce co-channel interference, so the reuse pattern can be **tightened**. Moving from a **7-cell to a 4-cell pattern nearly doubles capacity**, and **intracell reuse** of the same channel by spatial separation becomes possible [MA p9–10].

2. Multiple-antenna techniques (MIMO)

- **SU-MIMO multilayer transmission** sends **parallel data streams** between one Tx–Rx pair, for a **higher data rate in the same bandwidth**.
- **MU-MIMO** forms **directional beams to several users simultaneously** (spatial multiple access), giving **more users per cell**.
- Diversity and beamforming improve link quality and coverage. 4×4 and 8×8 MIMO are in use; massive MIMO is coming [WC-B 5.4, Fig 5.9].

3. Bandwidth expansion

- **Carrier aggregation**: combine channels (802.11n/ac: 20 → 40/80/160 MHz; LTE aggregates carriers).
- **Millimetre-wave bands (30–300 GHz)**: much wider channels are available (at the cost of obstruction and absorption problems). These are needed because Shannon limits the bps/Hz achievable for a given SNR [WC-B 5.4; WC-S 52].

*(Other valid choices: **efficient multiple access**: CDMA gives 4–5× the capacity of other digital systems [MA p8] and OFDMA allows tight subcarrier packing and opportunistic scheduling [OF]; **adaptive modulation and coding** [WC-S 51].)*

Q6. Explain the spread spectrum principles of FHSS and DSSS. (10)

Identical to T1A Q4 (Part 5).

Q7. Describe CDMA and explain why it is difficult to jam. (10)

Expanded version of Exam Q1(d) for 10 marks:

Description (6 marks):

1. **Spread-spectrum multiple access**: all users **share the entire bandwidth all the time**; there are no separate frequencies or time slots [MA p7; MX].
2. **Unique pseudorandom (orthogonal) code per user**, shared by the mobile and the base station [MA p8].
3. **Spreading**: data are XORed/multiplied with the code at a much higher **chip rate**. E.g. IS-95: **1.2288 Mchip/s** spreading 13 kbps voice over **1.25 MHz**, with up to **64 codes/users** per channel [AT].
4. **Despreading**: the receiver's **correlator** uses the same code. Only the wanted signal is despread and passes the **narrowband filter**; other users' signals remain noise-like [MA p8].
5. **Power control**: all users transmit at the minimum required power, to limit mutual interference (the near-far problem) and save battery [MA p8].
6. **Advantages**: greatest spectrum efficiency (8–10× analogue, 4–5× other digital), **soft handoff**, simplified frequency planning, privacy (Walsh codes). Used in IS-95 and 3G [MA p8].

Why it is difficult to jam (4 marks):

1. The signal occupies a **very wide band at a power density below the noise floor**. A jammer would need enormous power across the whole band.
2. **Processing gain**: despreading **spreads the jammer** over the wide band while concentrating the wanted signal. The narrowband filter then rejects most of the jamming energy (SUPPLEMENTARY: gain ≈ chip rate/bit rate, e.g. ≈ 21 dB for IS-95).
3. The jammer **doesn't know the pseudorandom code** (or hop sequence), so it cannot correlate with or mimic the signal. In FH, *"a malicious sender may jam one hopping period, but not the whole period"* [SS].

4. **Designed for anti-jamming**: spread spectrum's purposes include suppressing interference and making interception difficult [MA p7]. CDMA was a **military technology used in WWII to foil German jamming** [MX].

Q8. Describe the difference between OFDM and OFDMA. (10)

OFDM (4 marks): a **multicarrier modulation/transmission technique**. A high-rate stream (R bps) is split by a serial-to-parallel converter into **N lower-rate substreams (R/N)**, carried on **N orthogonal subcarriers** (spacing fb = a multiple of 1/T, e.g. 15 kHz in LTE). Orthogonality lets the spectra **overlap without interference** (≈ 6× more signals than FDM in the notes' example). Longer symbols (N·Ts) **defeat ISI** and **frequency-selective fading** hits only some subcarriers. **In OFDM all subcarriers carry the data of a single source** [OF §8.1].

OFDMA (4 marks): a **multiple access scheme built on OFDM**: a combination of **FDMA and TDMA**, where **different users are assigned subsets of the subcarriers (subchannels / LTE resource blocks of 12 subcarriers) at different times**. Subchannels can be formed from **adjacent, regularly spaced or randomly spaced** subcarriers, and are allocated by **opportunistic scheduling** (efficiency, proportional fairness $r_i/\bar{r}_i$, requirements, priority). In the uplink, a user **concentrates power** on its own subchannels, saving battery [OF §8.2; AT].

Key differences (2 marks):

	OFDM	OFDMA
Type	Modulation / transmission technique	Multiple access technique
Users per symbol	One user uses all subcarriers	Many users share the subcarriers simultaneously
Multiple access	Needs another method (e.g. **CSMA** in 802.11, TDMA)	Built in (frequency + time)
Examples	802.11a/g/n	**LTE downlink**, WiMAX

"Not all technologies that use OFDM use OFDMA. Some forms of 802.11 use OFDM for the signal but CSMA for multiple access" [OF §8.2].

Q9. What distinguishes single-user MIMO from multiple-user MIMO? (8)

MIMO uses multiple antennas at transmitter and receiver, for diversity, beamforming, multi-user MIMO and multilayer transmission [WC-B 5.4].

SU-MIMO (4 marks): all spatial streams go between the base station/AP and **one user** at a time. **Multilayer transmission** sends **multiple parallel data streams** over the same frequency and time between that single Tx–Rx pair. **Goal: a higher data rate for that user in a given bandwidth.** The user device needs **multiple antennas** (number of streams ≤ min(Tx, Rx antennas)).

MU-MIMO (4 marks): *"with enough MIMO antennas, directional antenna beams can be established to **multiple users simultaneously**"* [WC-B 5.4]. The BS **beamforms** separate streams to **different users** on the same time/frequency: **spatial division multiple access**. **Goal: higher system capacity, i.e. more users per cell.** Users can have single antennas, but the BS needs **many antennas** and **knowledge of each user's channel** to steer the beams and cancel inter-user interference.

Summary: SU-MIMO increases **one user's throughput**; MU-MIMO increases **the number of users served simultaneously**, i.e. cell capacity [WC-B Fig 5.9].

TEST 2 2025 (T2-2025; 100 marks)

Q1. State and explain the three categories of routing protocols in MANETs. (12)

Use Exam Q2(d) (Part 5), with 4 marks per category. For 12 marks, add to each category:

- **Proactive**: works like **distance-vector/link-state routing adapted to mobility**. DSDV adds **destination sequence numbers** to avoid loops and count-to-infinity, with full and incremental table dumps. **OLSR** uses multipoint relays to reduce flooding. **Best for** small networks with low mobility and frequent traffic.
- **Reactive**: **AODV**: RREQ broadcast → reverse path set up → RREP unicast back → forward route installed; **RERR** on link breaks; sequence numbers keep routes fresh [BT-S 87 for the outline]. **DSR**: the **whole source route** is carried in each packet; route caching. **Best for** large, highly mobile networks with occasional traffic.
- **Hybrid**: **ZRP**: each node has a **zone** (radius in hops). Intra-zone routing is proactive (IARP); inter-zone routing is reactive (IERP), querying only **border nodes** ("bordercasting"). Combines low latency nearby with low overhead far away.

Q2. Dijkstra on Figure 1 from x to all nodes. (8)

Step	N'	D(t),p	D(u),p	D(v),p	D(w),p	D(y),p	D(z),p
0	x	∞	∞	**3,x**	6,x	6,x	8,x
1	xv	7,v	**6,v**	—	6,x	6,x	8,x
2	xvu	7,v	—	—	**6,x**	6,x	8,x
3	xvuw	7,v	—	—	—	**6,x**	8,x
4	xvuwy	**7,v**	—	—	—	—	8,x
5	xvuwyt	—	—	—	—	—	**8,x**
6	xvuwytz	—	—	—	—	—	—

Updates: via v (3): t = 3 + 4 = 7; u = 3 + 3 = 6; w via v = 7 > 6 and y via v = 11 > 6 (no change). Ties at 6 (u, w, y) broken alphabetically. Via u: t via u = 8 > 7; w via u = 9 > 6. Via y: t = 13, z = 18 (no change).

To	v	u	w	y	t	z
Cost	3	6	6	6	7	8
Path	x-v	x-v-u	x-w	x-y	x-v-t	x-z
First hop	v	v	w	y	v	z

Q3. What is the count-to-infinity problem in distance-vector routing? (5)

Use the first paragraph of Exam Q3(d) (Part 5), plus the cause and the fix (5 marks):

- **Cause**: DV nodes know only their **neighbours' distance estimates**, not the paths behind them. After a link failure or cost increase, a node can adopt a neighbour's route that **goes back through itself**.
- **Effect**: a **routing loop**. Estimates rise by small steps with each exchange (**"bad news travels slowly"**), up to the true cost or to **infinity** if unreachable. Packets loop meanwhile. In the y/z/x example (4, 1, 50 → 60), it takes **44 iterations**.

- **Fixes**: **poisoned reverse** (advertise ∞ to the neighbour you route through), **split horizon**, a **maximum metric** (RIP: 16 = infinity).

Q4. Explain the hidden terminal problem. Does it occur in Bluetooth? (10)

Hidden terminal problem (6 marks): in a wireless LAN, station **B** is in range of both **A** and **C**, but **A and C cannot hear each other** (distance or an obstacle) [FZ Fig 15.3].

- A sends to B. C senses the medium, **hears nothing** and concludes it is idle, so it also transmits (to B). The two frames **collide at B**.
- Neither A nor C can **detect** the collision: wireless stations can't send and listen at once, and they can't hear each other anyway. So the whole frames are wasted, followed by ACK timeouts and retransmissions. **Capacity drops** [FZ].
- This is one of the three reasons **CSMA/CD can't be used** in wireless [WL 29].
- **Solution in 802.11**: the **RTS/CTS handshake**. The **CTS from B** (with the duration) reaches C, which sets its **NAV** and defers [FZ; WL 39, 43].

In Bluetooth (4 marks): **No, not within a piconet.** Access is **centrally scheduled by the master** (FH-TDD-TDMA polling). A slave transmits **only in the slot immediately after the master addresses it**, and SCO slots are **reserved** [BT-B]. Since no device transmits on its own after sensing the channel, two hidden slaves can never start transmitting at the same time to the master. (Collisions between **different piconets** happen only when they hop to the same frequency in the same slot. That is a separate FH-CDMA effect handled by FEC/ARQ [BT-B].)

Q5. Compare wired and wireless LANs. What unique concerns must the designer of a wireless LAN address? (10)

Comparison (5 marks): use the wired-vs-wireless table in Part 3 §9.3 [FZ]. Medium (wires vs broadcast air); hosts (fixed attachment vs free movement); isolated LAN (switch vs ad hoc); connection to other networks (router vs AP gluing wireless to wired); moving between them (only NICs and switch↔AP change; MAC addresses change, IP addresses stay the same); access method (CSMA/CD vs CSMA/CA); characteristics (wireless adds attenuation, interference, multipath and a higher error rate). Also from [WL 3]: wireless gives flexibility, ad hoc operation and disaster robustness, but lower bandwidth, lower QoS and higher delay.

Unique concerns (5 marks; 1 mark each, any five):

1. **Attenuation/path loss and coverage**: AP placement; more APs [WCP].
2. **Interference** in the crowded, unprotected **ISM band**: channel planning, 5 GHz [WL 23].
3. **Multipath and fading**: diversity, MIMO, OFDM, rate adaptation [WCP; WL 26].
4. **Higher error rates**: ACKs, retransmission, FEC, **fragmentation** [FZ; WL 27].
5. **Medium access without collision detection**: **hidden and exposed stations**, so CSMA/CA, RTS/CTS and NAV [FZ].
6. **Mobility/roaming** between BSSs and ESSs; association [FZ; WL 12].
7. **Power management** for battery devices [WL 4].
8. **Security and privacy** on a broadcast medium [WL 4, 27].
9. **Regulation**: licence-free but power-limited (≤ 1 W); national restrictions [WL 3, 23].
10. **QoS** for time-sensitive traffic: PCF, rate adaptation [WL 40].

Q6. A wireless receiving site has occasional loss of signal due to interference. Discuss the steps you would take. (10)

Identical to T1A Q12 (Part 5).

Q7. While Bluetooth's profiles add significant complexity, how can they benefit applications from the application's perspective? (10)

Exam Q4(c) answer, expanded to 10 marks. (2 marks per developed point.)

1. **Guaranteed interoperability between vendors**: the profile specification exists *"to define a standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A user's headset, car kit or keyboard works with any phone or PC supporting the same profile.
2. **A complete recipe for each usage model**: a profile is *"a set of protocols that implement a particular Bluetooth-based application"*. It states which parts of the core specification are **mandatory, optional and not applicable** [BT-B]. Application developers don't have to interpret the 2500+ page core themselves; the design decisions are made for them.
3. **Reuse of existing protocols and other profiles**: profiles build on each other. E.g. the **File Transfer Profile depends on the Generic Object Exchange, Serial Port and Generic Access profiles**, using **OBEX** over **RFCOMM** and **L2CAP** [BT-B]. Developers reuse proven components rather than writing new ones.
4. **Smaller, cheaper, lower-power implementations**: a device implements **only the profiles it needs** (a mouse doesn't need file transfer), keeping Bluetooth's key attributes of **low cost and low power** [BT-B].
5. **Consistent behaviour and easy setup**: every device supports the **Generic Access Profile**, so discovery, connection and security work the same everywhere. With **SDP**, devices can query each other's services and connect automatically. This gives plug-and-play operation and simpler testing and certification [BT-B].

Q8. Scatternet of 2 piconets, no CSMA, 80 frequencies: probability of collision? (10)

Exam Q4(e) answer, written out in full for 10 marks:

Model and assumptions (3 marks):

- Each piconet's master sets a **pseudorandom hop sequence** from its own address, so the two piconets hop **independently** [BT-B].
- **80** equally likely frequencies; one hop per slot.
- Both piconets **always transmit** (always have packets), with **no carrier sensing**.
- A **collision** occurs when both use the **same frequency in the same slot**.
- A **successful transmission** = poll (master → slave) **and** reply (slave → master), so **2 slots on 2 different hops**.

Calculation (5 marks):

1. P(same frequency in a given slot) = 80 × (1/80)(1/80) = **1/80 = 0.0125**
2. P(no collision in a slot) = 1 − 1/80 = **79/80 = 0.9875**
3. The two slots are independent, so P(poll AND reply both collision-free) = $(79/80)^2$ = **0.97516**
4. **P(collision during the transaction) = $1 - (79/80)^2 = 159/6400 \approx 0.0248$**

Answer (2 marks): **about 2.5%** of poll/response transactions suffer a collision (1.25% for any single packet). Performance degrades as more piconets are added, as the notes say: *"as the number of*

piconets in an area increases, the number of collisions increases" [BT-B]. SUPPLEMENTARY: with k other piconets, P(slot OK) = $(79/80)^k$.

Q9. A Bluetooth device can be in two piconets at the same time. Is there any reason one device cannot be the controller in both at the same time? (5)

Identical to Exam Q4(d) (Part 5), plus one more mark: a device in two piconets must **time-share**. It follows one piconet's hop sequence and timing, then the other's. As master it would have to **impose its own hop sequence on both**, which would merge them into one.

Q10. Two ISPs in a café. (i) Both on channel 11. (10) (ii) Channels 1 and 11. (10)

Use Exam Q5(e) (Part 5), expanded: **(i) (10 marks)**

- **Does it break down?** No. 802.11's **CSMA/CA is distributed**: every station senses the channel before transmitting, whatever its ISP or AP. The two BSSs simply become **one shared collision domain** on channel 11 [WL 32–35].
- **Normal operation**: when a station of ISP 1 is transmitting, stations of ISP 2 that hear it (or its RTS/CTS) **defer**. Their **NAV** is set from the duration fields, so the channel is **time-shared** between the two BSSs. Each AP only processes frames carrying **its own BSSID**; the other ISP's frames are ignored after reception. The separate IP address blocks are irrelevant at the MAC layer.
- **Simultaneous transmission by two stations** (one per ISP): if both finish DIFS + backoff in the same slot, or they are **hidden** from each other, their frames **collide** at the receiving AP(s). With no collision detection, both frames are lost; **no ACK** returns; each station assumes a collision, **doubles its contention window**, picks a new random backoff and **retransmits** [WL 33, 35, 38].
- **Consequences**: more collisions, retransmissions and backoff delay; the channel's capacity is **shared**, so each ISP gets roughly **half the throughput**; **co-channel interference** raises the error rate [WL 12; WCP].

(ii) (10 marks)

- Channel 1 (2.412 GHz) and channel 11 (2.462 GHz) are **50 MHz apart**. Each ~22 MHz-wide signal [WC-S 4] does not overlap the other: they are **non-overlapping channels** (SUPPLEMENTARY).
- The two BSSs are now **independent collision domains**. A station's carrier sense on channel 1 does **not** detect transmissions on channel 11, and vice versa.
- **Two stations of different ISPs transmitting at the same time do not collide.** Both frames are received correctly by their own APs and ACKed.
- Each ISP gets the **full capacity** of its own channel, so aggregate throughput roughly **doubles**, with fewer collisions, backoffs and retransmissions and **no co-channel interference** between the networks.
- Stations of the same ISP still contend with each other as normal under CSMA/CA.

COM5231 TEST 2 2026 (T2-2026; 50 marks)

Q1. Distinguish between routing, packet forwarding and switching. For each, state its purpose and the information used to make its decision. (6)

SUPPLEMENTARY (2 marks each).

- **Routing**: **purpose**: determine the **end-to-end path** that packets take from source to destination across the network (a network-wide, control-plane function performed by routing algorithms such as Dijkstra/link-state and Bellman-Ford/distance-vector). **Information used**: the **network topology and link costs**, learned by exchanging routing messages with other routers. The output populates the **routing/forwarding table**.
- **Packet forwarding**: **purpose**: inside one router, **move each arriving packet from its input port to the correct output port/link** (a local, per-packet, data-plane action). **Information used**: the packet's **destination IP (network-layer) address**, looked up in the **forwarding table** (longest-prefix match).
- **Switching**: **purpose**: move **frames** to the correct outgoing port **within a LAN** at the link layer (also the term for transferring packets across a router's internal switching fabric). **Information used**: the frame's **destination MAC address**, looked up in the **switch table**, which the switch builds by **self-learning** from the source MAC addresses of arriving frames.

Q2. Data byte 11001010; even-parity Hamming code; positions numbered from 1 at the left; check bits at 1, 2, 4, 8; data bits left to right in the remaining positions; no overall parity bit.

(a) Minimum number of check bits; show the inequality and verify. (2)

Inequality: **$2^r \geq m + r + 1$**, with m = 8 data bits.

- r = 3: $2^3 = 8 \geq 8 + 3 + 1 = 12$? **No**.
- r = 4: $2^4 = 16 \geq 8 + 4 + 1 = 13$? **Yes**. **Minimum r = 4** check bits, giving a 12-bit codeword.

(b) Data-bit placement and value of each check bit, with the positions covered. (4)

Position	1	2	3	4	5	6	7	8	9	10	11	12
Bit	P1	P2	**1**	P4	**1**	**0**	**0**	P8	**1**	**0**	**1**	**0**

Check bit	Positions covered	Data bits	Number of 1s	Even-parity value
P1	1, 3, 5, 7, 9, 11	1, 1, 0, 1, 1	4	**0**
P2	2, 3, 6, 7, 10, 11	1, 0, 0, 0, 1	2	**0**
P4	4, 5, 6, 7, 12	1, 0, 0, 0	1	**1**
P8	8, 9, 10, 11, 12	1, 0, 1, 0	2	**0**

(c) Complete transmitted codeword (positions 1 → 12). (2)

0 0 1 1 1 0 0 0 1 0 1 0 → 001110001010

Q3. CRC: data 110101, generator 1011 ($G(x) = x^3 + x + 1$), modulo-2 division.

(a) Number of zeros appended and the resulting bit string. (1)

Degree 3 → **append 3 zeros** → **110101000**.

(b) Show the division and the CRC remainder. (3)

Step	Window	⊕ 1011	Result	Working value
1	1101	1011	0110	011001000
2	1100	1011	0111	001111000
3	1111	1011	0100	000100000
4	1000	1011	0011	000001100
5	1100	1011	0111	000000111

```
110101000
1011
----
 1100
 1011
 ----
  1111
  1011
  ----
   1000
   1011
   ----
     1100
     1011
     ----
      111   ← remainder
```

CRC remainder = 111.

(c) Complete transmitted bit string. (2)

110101111 (data 110101 followed by CRC 111). Check: 110101111 ÷ 1011 → remainder 000 ✔.

Q4. Simplified Internet checksum, 4-bit words. Message 1010 1100 0110 1001; one's-complement addition with end-around carry.

(a) Show the addition and calculate the checksum. (3)

Step	Addition	Raw result	End-around carry	Running sum
1	1010 + 1100	1 0110	0110 + 1	**0111**
2	0111 + 0110	1101	none	**1101**
3	1101 + 1001	1 0110	0110 + 1	**0111**

Sum = 0111. Checksum = one's complement = 1000. (Decimal check: 10 + 12 + 6 + 9 = 37; mod 15 → 7 = 0111 ✔.)

(b) 4-bit result expected when the receiver adds all message words and the checksum, assuming no error. (1)

0111 + 1000 = **1111** (all ones; complementing it gives 0000, meaning no error detected).

Q5. Define a single-bit error and a burst error. Explain how the length of a burst error is measured. (3)

- **Single-bit error**: only **one bit** of the data unit changes (0 → 1 or 1 → 0). (1)

- **Burst error**: **two or more bits** in the data unit change (not necessarily consecutive). (1)
- **Burst length**: measured from the **first corrupted bit to the last corrupted bit, inclusive**. Bits between them may or may not be corrupted. E.g. errors at bit positions 3 and 6 → burst length 6 − 3 + 1 = **4**. (1)

Q6. Even-parity scheme with one parity bit per block. Is it guaranteed to detect each of the following? Justify.

(a) A single-bit error. (2)

Yes, always. Even parity makes the number of 1s even. Flipping **one** bit changes the count by exactly one, making it **odd**, so the receiver's check fails and the error is detected.

(b) A burst error. (2)

No, not guaranteed. Parity detects only an **odd** number of bit errors. A burst that flips an **even** number of bits (2, 4, …) leaves the count of 1s even, so the block **passes** undetected. E.g. 1011 0010 → 1**10**1 0010: two errors, still four 1s. A burst is detected only if it happens to flip an odd number of bits.

(c) Can a single parity bit identify and correct the erroneous bit? (1)

No. It only shows that **an odd number of bits** changed, not **which** bit. Correction needs more redundancy (e.g. Hamming code or 2-D parity) or retransmission.

Q7. Bluetooth profiles add complexity. Explain two ways in which profiles benefit application developers or users. (4)

1. **Interoperability (users)** (2): profiles define a *"standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A user can pair any compliant headset, car kit or keyboard with any phone without compatibility worries.
2. **Ready-made, reusable implementation (developers)** (2): a profile specifies **exactly which protocols and features are mandatory, optional or not applicable** for a usage model, and builds on other profiles. E.g. the File Transfer Profile reuses OBEX, RFCOMM, L2CAP and the Generic Access Profile [BT-B]. Developers implement a known recipe (and only the subset they need), which saves effort and keeps devices small, cheap and low-power.

Q8. Compare the Bluetooth Classic central (master) with an access point in an 802.11 infrastructure WLAN. Explain one difference in medium-access coordination and one in their roles in connecting devices to other networks. (4)

- **Medium-access coordination (2)**: the **Bluetooth central schedules all access**. It defines the hop sequence and timing, and **polls** slaves, who may transmit **only in the slot after being addressed** (FH-TDD-TDMA), so there is **no contention or collision inside the piconet** [BT-B]. In **802.11**, stations **contend using CSMA/CA (DCF)**: sense, DIFS, random backoff, RTS/CTS, NAV. **Collisions are possible**, and the AP does not normally schedule access (polling via PCF is optional) [WL 32–40].
- **Connecting to other networks (2)**: the **802.11 AP is a bridge/portal**: it *"provides a link to other networks and acts as a relay point for all MAC frames"*, connecting the BSS to the wired distribution system/Internet [BT-S 64; FZ]. The **Bluetooth central only sets up the piconet and relays traffic within it** [BT-S 64]. Slaves can reach other piconets only via **scatternet bridge devices**, and a piconet is a short-range PAN, not an infrastructure gateway.

Q9. Dijkstra's shortest-path algorithm on the given network.

Link costs read from the figure: A–B 2, A–G 6, B–C 7, B–E 2, E–G 1, E–F 2, F–C 3, F–H 2, G–H 4, C–D 3, H–D 2.

(a) Shortest path from A to all nodes (step by step). (6)

Step	N'	B	C	D	E	F	G	H
0	A	**2,A**	∞	∞	∞	∞	6,A	∞
1	AB	—	9,B	∞	**4,B**	∞	6,A	∞
2	ABE	—	9,B	∞	—	6,E	**5,E**	∞
3	ABEG	—	9,B	∞	—	**6,E**	—	9,G
4	ABEGF	—	9,B	∞	—	—	—	**8,F**
5	ABEGFH	—	**9,B**	10,H	—	—	—	—
6	ABEGFHC	—	—	**10,H**	—	—	—	—
7	ABEGFHCD	—	—	—	—	—	—	—

Working:

- Step 0: A's neighbours are B 2 and G 6. Smallest = **B**.
- Step 1: via B (2): C = 2 + 7 = 9; E = 2 + 2 = 4. Smallest = **E (4)**.
- Step 2: via E (4): G = 4 + 1 = **5** (< 6, update); F = 4 + 2 = 6. Smallest = **G (5)**.
- Step 3: via G (5): H = 5 + 4 = 9. Smallest = **F (6)**.
- Step 4: via F (6): C = 6 + 3 = 9 (equal to 9, keep B); H = 6 + 2 = **8** (< 9, update). Smallest = **H (8)**.
- Step 5: via H (8): D = 8 + 2 = 10. Smallest = **C (9)**.
- Step 6: via C (9): D = 9 + 3 = 12 > 10 (no change). Smallest = **D (10)**. Done.

(b) Final shortest-path cost and complete route from A to B, C, D, E and F. (4)

Destination	Cost	Route
B	**2**	A → B
C	**9**	A → B → C *(equal-cost alternative A → B → E → F → C = 9)*
D	**10**	A → B → E → F → H → D
E	**4**	A → B → E
F	**6**	A → B → E → F

(Also: G = 5 via A → B → E → G; H = 8 via A → B → E → F → H.)

BLUETOOTH EXERCISES (20 questions) [BTX]

1. Why is Bluetooth called a WPAN, not a WLAN? Its range is ~**10 m** (a person's personal space), it targets **low power, low cost cable replacement** between a user's own devices, and a piconet holds only **8 active devices**. A WLAN covers a building or campus, serves many users and connects to wider networks [BT-B; BT-S 56].

2. Alternate name for Bluetooth Smart, and which version? Bluetooth Low Energy (BLE), introduced in **Bluetooth 4.0** [BT-S 53].

3. How does Bluetooth 3.0 reach 24 Mbps? By the **Alternative MAC/PHY (AMP)**: Bluetooth 3.0 + HS uses a co-located **802.11 (Wi-Fi) link** for high-rate data. The Bluetooth radio still does discovery, association, connection setup, profile configuration and maintenance; a **Protocol Adaptation Layer (PAL)** maps Bluetooth protocols onto the 802.11 MAC/PHY [BT-S 52; BT-B].

4. Compare a piconet and a scatternet. Piconet: the basic unit, **1 master + up to 7 active slaves**, sharing **one hopping sequence** set by the master; slaves talk only to the master. **Scatternet**: **two or more overlapping piconets** linked by devices that belong to more than one (as slave in both, or master in one and slave in the other). **The piconet shares the logical channel and data transfer; the scatternet shares the physical area and total bandwidth** (FH-CDMA) [BT-B].

5. Can a piconet have more than eight stations? Not more than eight active ones: 1 master + **7 active slaves**, because the **AM_ADDR** field is **3 bits** (8 values, with 0 reserved for broadcast) [BT-S header]. SUPPLEMENTARY: more devices can be **parked** (synchronised but inactive) and swapped in.

6. Relationship between master and slave? The **master determines the hopping sequence (channel) and phase (timing)** from its own address. **Slaves tune to it**, may **communicate only with the master**, and **only when granted permission** (polled) [BT-B].

7. Types of links between master and slave. SCO (synchronous connection-oriented: reserved symmetric slots, fixed bandwidth, **no retransmission**, used for voice); **eSCO** (extended SCO: reserved slots, **asymmetric** allowed, **retransmission** supported); **ACL** (asynchronous connectionless: **point-to-multipoint**, packet-switched, uses unreserved slots, **retransmission**, polled) [BT-B].

8. Master vs base station in 802.11. See Exam Q4(a) (Part 5) and T2-2026 Q8.

9. Can one device be controller of two piconets? No. See Exam Q4(d) (Part 5).

10. Role of the radio layer? It specifies the **air interface**: the 2.4 GHz ISM band, **79 × 1 MHz channels**, **frequency hopping**, **GFSK** (π/4-DQPSK/8DPSK for EDR) modulation, and **transmit power classes 1–3** [BT-B].

11. Actual bandwidth used? At any instant a piconet uses **1 MHz** (one hop channel). Over time it hops across the **79 MHz (≈ 80 MHz)** band, so the total spread bandwidth is ~79 MHz [BT-B].

12. Spread spectrum technique? Frequency hopping spread spectrum (FHSS), **1600 hops/s** (0.625 ms slots), with optional **adaptive frequency hopping** [BT-B].

13. Modulation technique in the radio layer? Gaussian frequency-shift keying (GFSK): binary 1 = **positive** frequency deviation, 0 = **negative** deviation from the carrier; 1 Msymbol/s × 1 bit = **1 Mbps**. EDR uses **π/4-DQPSK (2 bits/symbol, 2 Mbps)** and **8DPSK (3 bits/symbol, 3 Mbps)** [BT-B].

14. How can FH and TDD be combined? Each **625 µs slot** uses the **next frequency** in the hop sequence (FH), **and** the direction alternates between slots: master → slave, then slave → master (TDD). Multislot packets (3 or 5 slots) stay on one frequency, then jump to the frequency the sequence requires. This gives FH-TDD-TDMA [BT-B Fig 12.4–12.5].

15. How does FH-CDMA differ from DS-CDMA? FH-CDMA: users are separated by **different hopping sequences**. Each uses a narrow channel at a time, and collisions happen only when two users land on the same frequency in the same slot (Bluetooth scatternets). **DS-CDMA**: all users transmit **simultaneously over the whole wide band**, spread by **different chip codes**, and separated by **correlation** at the receiver (IS-95, WCDMA) [BT-B; MA; AT].

16. Do Bluetooth devices and microwave ovens interfere? Yes, they can. Microwave ovens radiate around **2.45 GHz**, inside Bluetooth's 2.4 GHz ISM band [WCP §10]. The effect is **limited**: Bluetooth hops over 79 channels 1600 times/s, so only hops near the oven's frequency suffer, **adaptive frequency hopping** can avoid bad channels, and lost packets are recovered by **FEC/ARQ** [BT-B].

17. MAC protocol in the baseband? TDMA with master-controlled polling over FH and TDD (FH-TDD-TDMA). There is no contention within a piconet; slaves transmit only when addressed [BT-B].

18. Role of L2CAP? It **adapts upper-layer protocols to the baseband**, providing **connectionless and connection-oriented** services. SUPPLEMENTARY detail: protocol multiplexing, segmentation and reassembly of large packets, and QoS [BT-B].

19. How are profiles an advantage to applications? See Exam Q4(c) and T2-2025 Q7.

20. Scatternet collision probability. See Exam Q4(e) and T2-2025 Q8: **$\approx 1 - (79/80)^2 \approx 2.5\%$**.

ERROR-DETECTION WORKSHEET EXERCISES [ED]

W1. Message 11001001, CRC $x^3 + 1$ (1001). Remainder **011** → transmit **11001001011**. Leftmost bit flipped → received 01001001011 → remainder **010** → non-zero → error detected. *(Worked in the worksheet; verified correct.)*

W2. CRC-8 ($x^8 + x^2 + x + 1$ = 100000111) on 1011 0011 0101 0110. Remainder **11010101** → transmit **1011 0011 0101 0110 1101 0101**. **7th bit flipped** → remainder **00101010** (101010) → error detected. *(Worksheet values verified.)*

W3. G = 10011, D = 1010101010: R? R = 0100. (a) 1000100101 → **1000**; (b) 0101101010 → **1111**; (c) 0110100011 → **1110**. *(Full division in Part 2, §8.8 CR5.)*

W4. Internet checksum of 1001 1100 1010 0011 (4-bit). 1011. (Exam Q2(c).)

W5. Hamming for 10101111: check bits and codeword. 4 check bits; codeword 101001001111. (Exam Q2(a).)

W6. M(x) = 110001110110, G(x) = $x^5 + x^2 + x + 1$: frame T(x)? Why modulo-2? G = **100111**, append 5 zeros, remainder **10101** → **T(x) = 110001110110 10101**. *(Division in Part 2, §8.8 CR6.)* **Why modulo-2**: addition and subtraction become the **same XOR operation with no carries or borrows**. That makes CRC computation trivial in hardware (**shift register + XOR gates**) and corresponds to **polynomial arithmetic over GF(2)**, which gives CRC its guaranteed detection properties.

W7. Single-bit vs burst errors and parity. See T1B Q1.